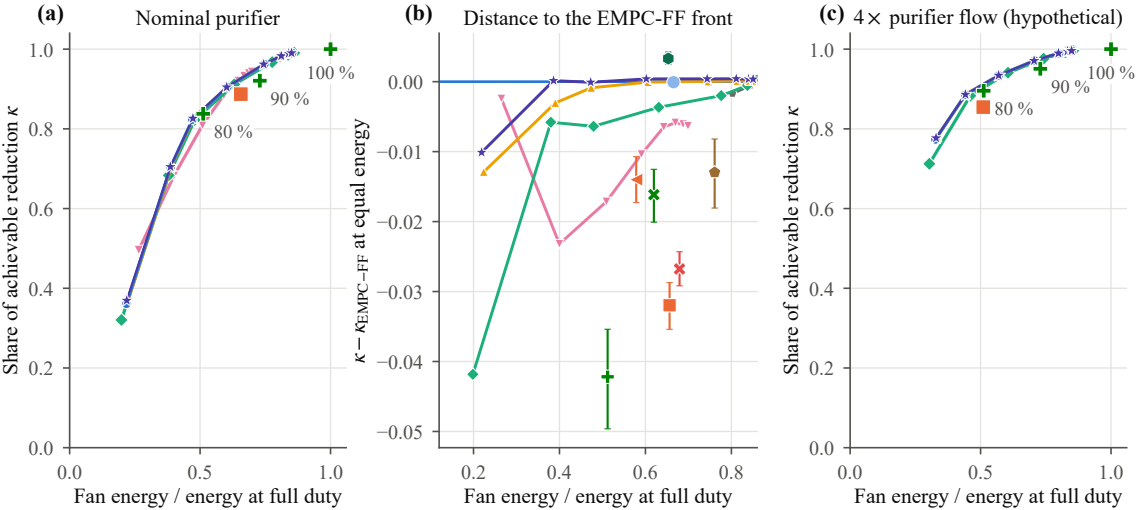

(a)
Nominal purifier
Share of achievable reduction κ
Fan energy / energy at full duty
100 %
90 %
80 %
(b)
Distance to the EMPC-FF front
κ − κEMPC−FF at equal energy
Fan energy / energy at full duty
(c)
4× purifier flow (hypothetical)
Share of achievable reduction κ
Fan energy / energy at full duty
100 %
90 %
80 %
Zone-tracking MPC (tuned)
Threshold + hysteresis (tuned)
Stationary costate law (CL-s)
EMPC, feedback only
EMPC, schedule feedforward
EMPC, oracle
Relay (baseline)
Constant duty 80/90/100 %
Relay, occupancy-gated
Scheduled duty (tuned)
Timer + relay (tuned)
PI, sigma-delta (tuned)
Rollout costate law (CL-r)
EMPC, relax-and-round